



NHAI 6TH INNOVATION HACKATHON

PROPOSAL

Project Name : RetroFusion AI

Tagline - Transforming Highway Safety Through AI-Powered Retroreflectivity Intelligence

Short Description : Smart Vision System for Real-Time Highway Reflectivity Monitoring

Solution Category	AI + Hardware Hybrid System
Applicable Highway Types	2-lane, 4-lane, 6-lane, 8-lane Expressways
Operating Conditions	Day / Night Dry / Wet Foggy With/Without Streetlight
Submission Date	23 April 2026
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1. Executive Summary

Every year, thousands of highway accidents happen in low-light conditions - not because drivers are reckless, but because the roads are invisible. RetroFusion AI changes that.

1.1. Problem (Unsafe, Slow Manual Measurement)

India's 1.45 lakh km of National Highway network is a circulatory system for its economy -yet each night, degraded retroreflective road markings and signs silently endanger millions of commuters.

Currently, retroreflectivity on highways is measured using handheld devices, which require technicians to physically stand on active, high-speed roads. This makes the process highly unsafe, exposing personnel to accident risks. Additionally, the method is extremely slow, allowing only limited measurements per hour, making it impractical for India's vast highway network. It also lacks continuous monitoring and performs poorly under night, fog, or wet conditions, leading to undetected degradation of road markings and signs, which directly impacts road safety. The current method of measuring retro-reflectivity uses handheld devices that require technicians to stand on live, high-speed traffic lanes, a procedure that is slow, dangerous, and utterly unscalable across the breadth of NHAI's network.

1.2. OUR Solution

RetroFusion AI is a **intelligent system** that measures retroreflectivity in real time at highway speeds using a combination of AI-based vision, photometric sensing, and calibration techniques. It automatically detects road assets, calculates reflectivity (RA values), and provides instant alerts with GPS-based reporting.

1.3. Key Innovation (AI + Hardware)

The key innovation lies in a **hybrid multi-modal fusion system** that combines **AI-based computer vision, photometric sensing with controlled IR illumination, and retroreflectometer calibration** into a single platform. The AI module detects and classifies road assets in real time, while sensors accurately measure reflected light under all conditions, and calibration ensures standard-compliant RA values. These inputs are integrated using **sensor fusion techniques** to deliver high accuracy and reliability. By unifying **vehicle-mounted measurement (Option i)** and **AI/ML estimation (Option ii)** into a single system, the solution achieves the **combined approach (Option iii)**, making it both technically advanced and practically deployable.

1.4. Impact (Speed, Safety, Scalability)

The system delivers significant real-world impact. In terms of speed, it performs continuous measurements at highway speeds, making it thousands of times faster than manual methods. For safety, it completely eliminates the need for personnel to be on the road, reducing risks to zero. In terms of scalability, the low-cost and modular design allows deployment across the entire national highway network. Additionally, it generates real-time data, GPS mapping, and predictive maintenance insights, enabling authorities to shift from reactive to proactive road safety management.

2. The Problem Statement

Current Scenario - Road safety is not just an engineering challenge - it is a moral imperative. India records over 1.5 lakh road fatalities annually. A significant proportion occur at night, on highways where degraded retroreflective materials fail to guide drivers. NHAI has recognized this and mandates minimum retro-reflectivity standards - but measurement has remained the weakest link in enforcement.

GAPS:

- No continuous monitoring
- No night/fog coverage
- No data intelligence

The Three Critical Failures of Today's System

Table 2.1: Comparison of current method and its impact

Failure Point	Current Method	Real-World Impact
Safety Risk	Technicians stand on live highway lanes	High accident risk to personnel
Speed	12–20 measurements per hour by hand	Impractical for 1.45 lakh km network
Coverage	Random spot checks only	Degraded signs go undetected for months
Weather Dependency	Mostly daytime, dry conditions only	Night/fog conditions unmeasured
Data	Paper logs, no GPS tagging	No actionable maintenance intelligence

A road marking that reflects even **10% below the minimum standard** can become nearly invisible to a driver traveling at 100 km/h, significantly increasing the risk of accidents. The current measurement methods are too slow and limited to detect such critical deficiencies in time.

Therefore, there is an **urgent need for an efficient, reliable, and scalable system** that ensures both road safety and regulatory compliance. It is essential to develop a solution that is easy to deploy across large highway networks. This need has motivated us in the development of **RetroFusion AI**, an intelligent system designed to address these challenges effectively.

3. Our Solution — RetroFusion AI

Our Proposed System

RetroFusion AI is a compact, mountable hardware-software system that transforms any inspection vehicle into a smart retro-reflectivity scanner. It operates continuously while the vehicle moves at highway speed, requiring zero manual intervention on the road surface.

3.1. Core System Architecture

The system is built on three integrated pillars:

- **PILLAR 1 — AI Vision Engine:** The Raspberry Pi HQ Camera (IMX477, 12.3MP) streams high-resolution imagery. A custom-trained neural network (lightweight CNN optimised for edge inference) detects road signs, pavement markings, road studs, and delineators in real time, classifying them by type and extracting their reflective properties from pixel luminance analysis.
- **PILLAR 2 — Photometric Sensor Fusion:** The TSL2591 high-dynamic-range lux sensor (188 dB dynamic range), paired with a controlled 850nm IR LED illuminator, measures incident and returned light intensity. This photometric data is processed using the ASTM E1745 retroreflectivity model to compute an estimated RA value ($\text{mcd}/\text{m}^2/\text{lux}$).
- **PILLAR 3 — Ground Truth Validation:** A portable retroreflectometer is integrated via USB, providing periodic calibration measurements. A sensor fusion algorithm (Kalman filter-based) merges AI estimates with sensor readings and retroreflectometer ground truth to produce a high-confidence final RA value.

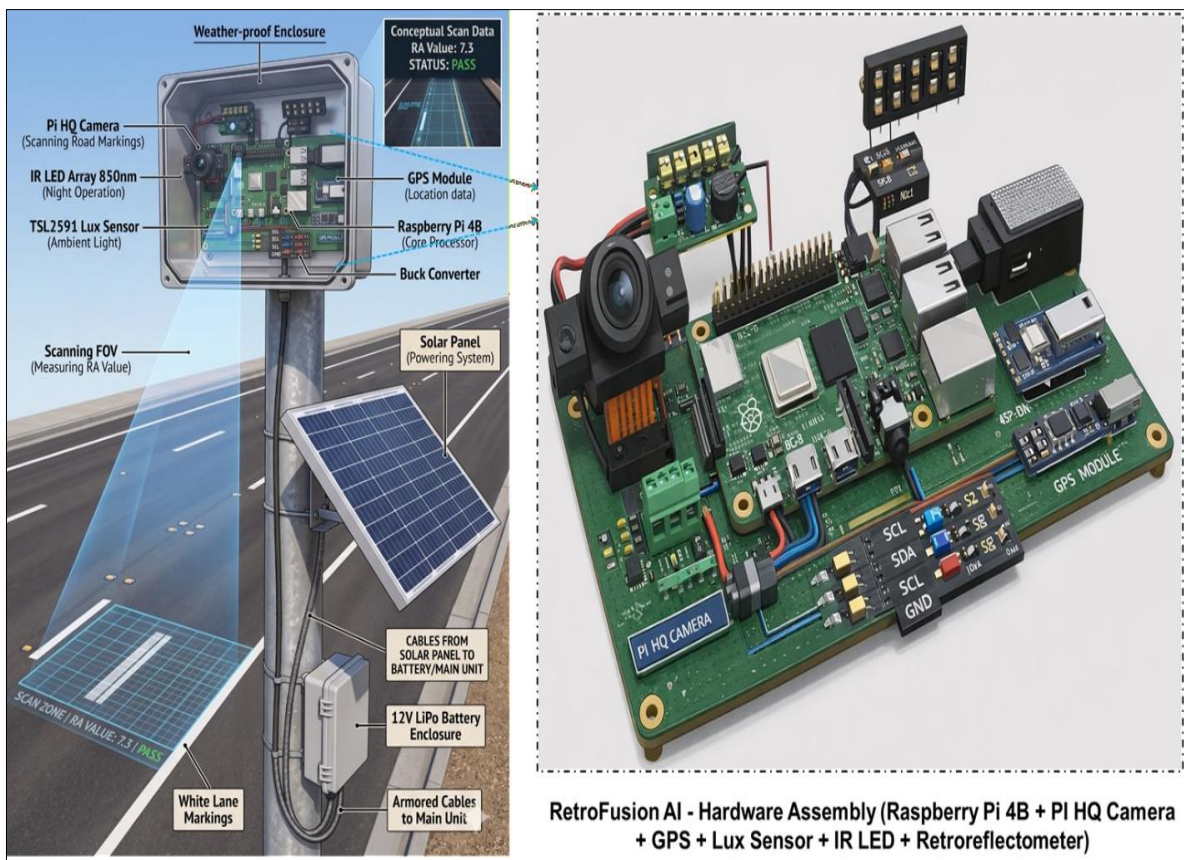


Fig 3.1: 3D view and RetroFusion AI - Hardware Assembly (Raspberry Pi 4B + PI HQ Camera + GPS + Lux Sensor + IR LED + Retroreflectometer)

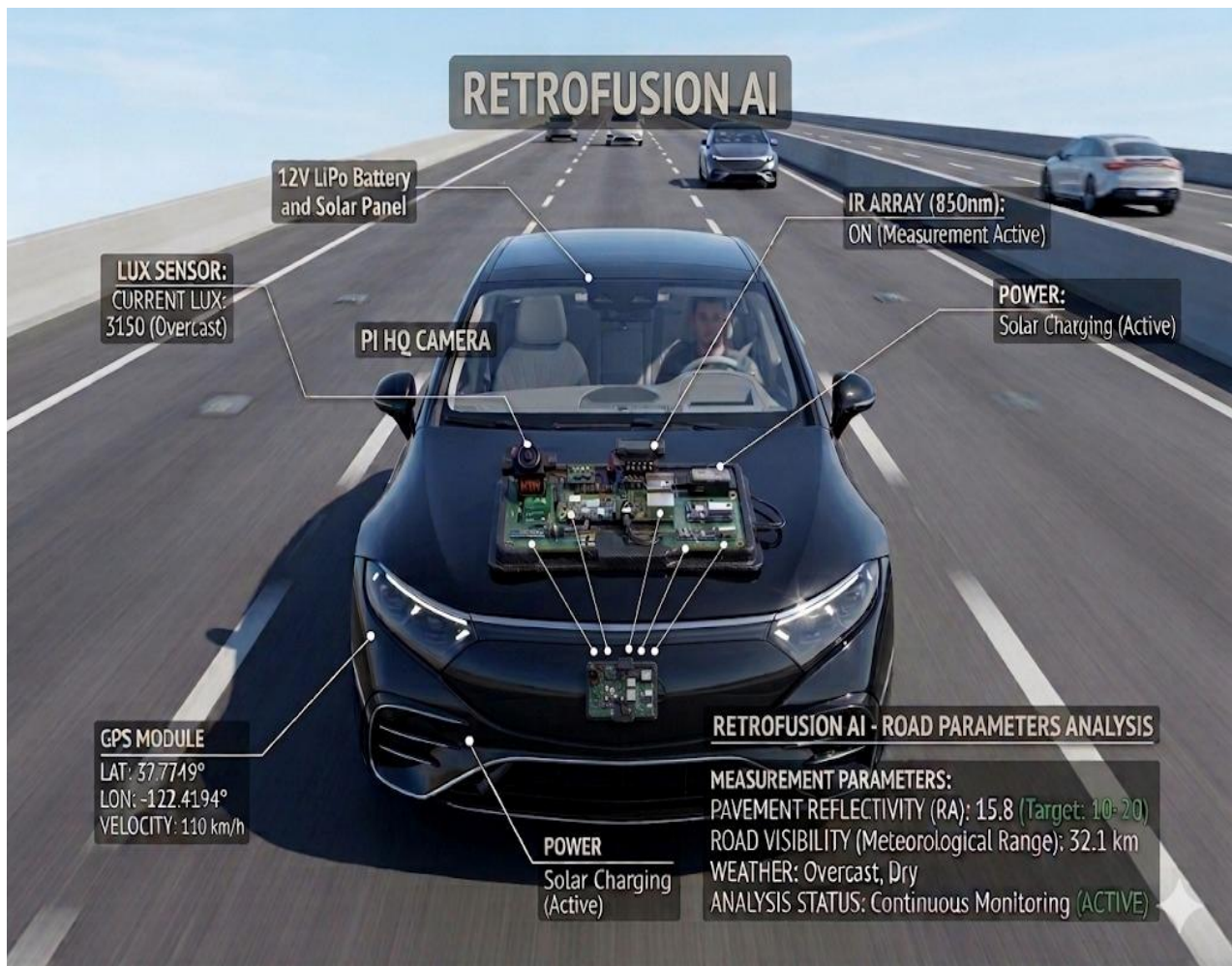


Fig 3.2: Device mounted(internal body) on a vehicle

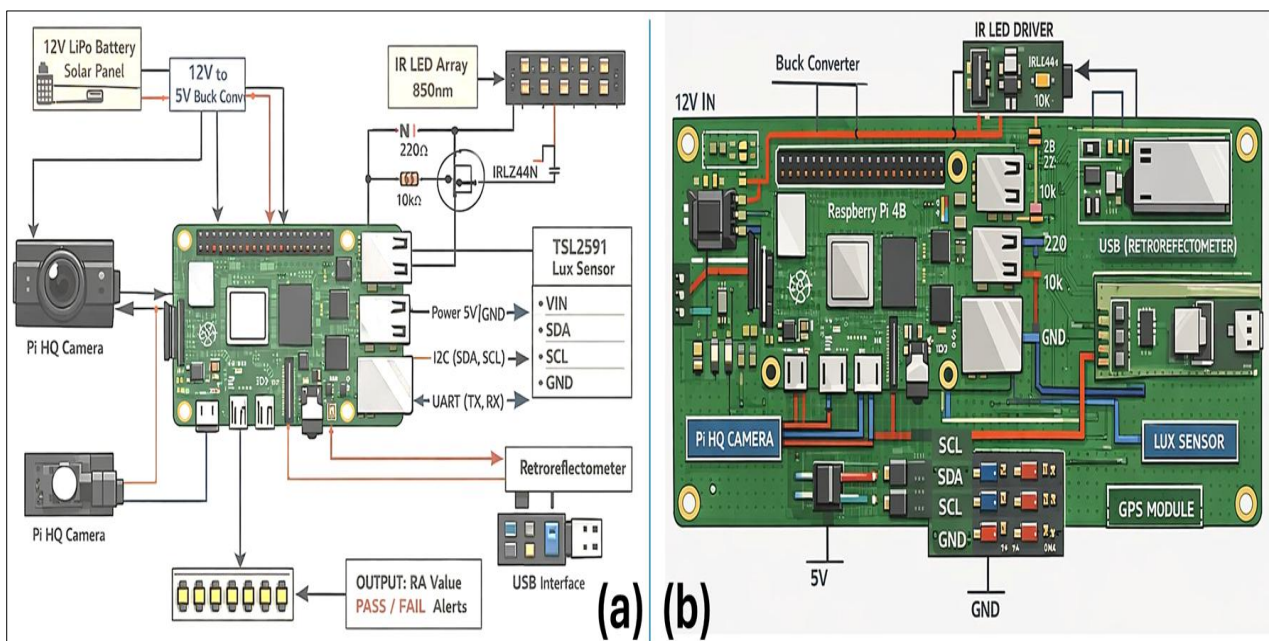


Fig 3.3: RetroFusion AI (a) Circuit Schematic & (b) PCB Layout architecture

3.2. How It Works: Step-by-Step

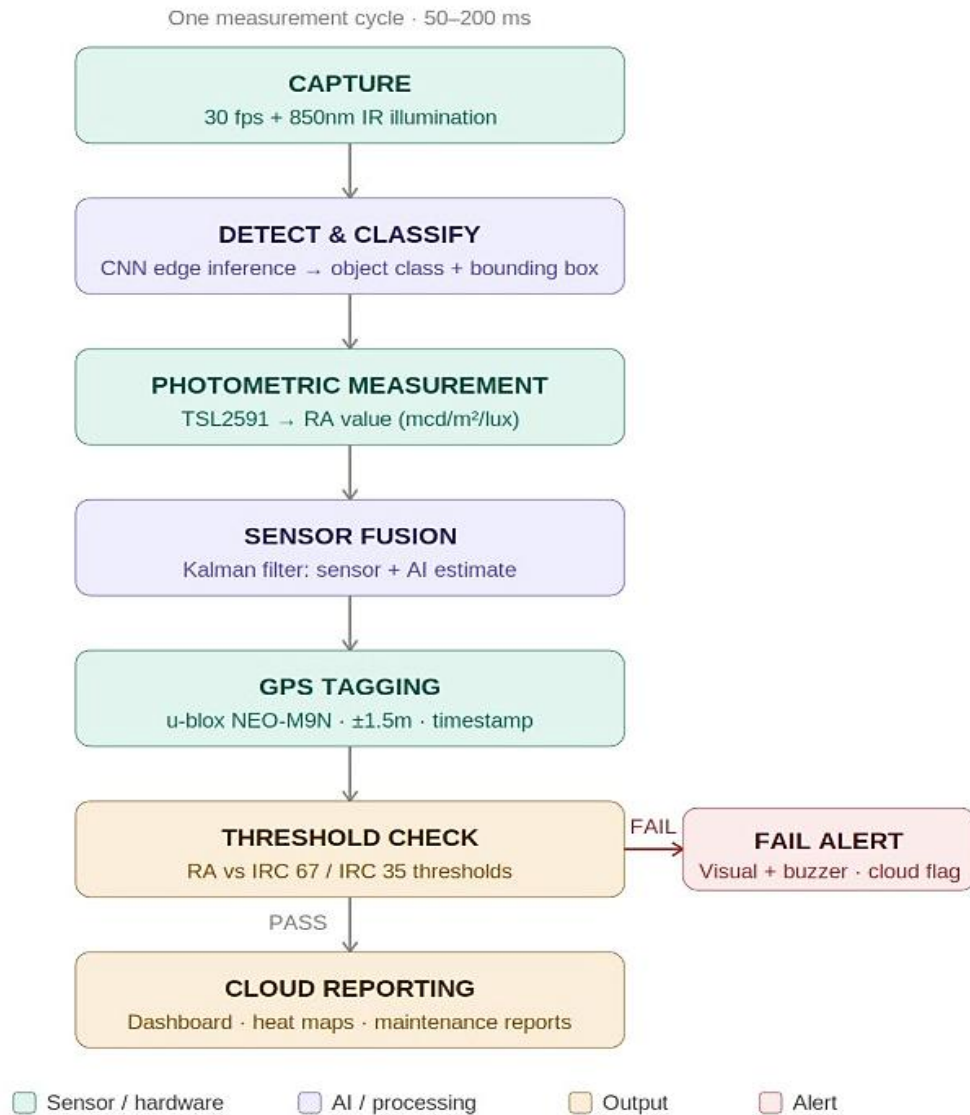


Fig 3.4: Flowchart of RetroFusion AI core architecture

The following workflow as in FIG 3.3 describes a single measurement cycle, occurring every 50–200ms as the vehicle travels:

1. **CAPTURE:** The Pi HQ Camera continuously captures frames at 30 fps. Simultaneously, the IR LED array illuminates the road ahead at 850nm, invisible to oncoming drivers.
2. **DETECT & CLASSIFY:** The AI model runs frame inference, detecting retroreflective objects — road markings (longitudinal lines, zebra crossings), traffic signs (gantry, shoulder-mounted), road studs, and delineators. Bounding boxes and object class are extracted per frame.
3. **PHOTOMETRIC MEASUREMENT:** The TSL2591 sensor measures the reflected IR light intensity in real time. Combined with known emitter power and geometry, the system computes an estimated luminance coefficient (RA value in mcd/m²/lux).

4. **SENSOR FUSION:** The Kalman filter algorithm merges AI-based luminance estimation with the TSL2591 sensor reading. Where retroreflectometer data is available (from periodic manual spot checks), it is used as a calibration anchor to eliminate drift.
5. **GPS TAGGING:** Each measurement is tagged with precise geolocation data from the u-blox NEO-M9N GPS module ($\pm 1.5\text{m}$ accuracy) and timestamped.
6. **OUTPUT & ALERT:** RA values are compared against IRC 67 and IRC 35 minimum thresholds. Locations falling below standard trigger a FAIL alert (visual + optional buzzer). All data streams to the cloud dashboard in real time via 4G/WiFi.
7. **REPORTING:** The web dashboard generates automated maintenance reports, heat maps of degraded zones, and predictive replacement timelines enabling NHAI project managers to prioritize interventions.

3.3 What Our System Measures :

- The system measures the **retroreflectivity (RA values)** of road assets including pavement markings, traffic signs, road studs, and delineators.
- It evaluates how effectively these elements reflect light back to drivers under real-world conditions such as **night driving, wet surfaces, and low visibility environments**.
- In addition to RA values, the system also assesses **visibility performance**, identifying faded, damaged, or non-compliant markings that fall below prescribed standards.

3.4. Our Proposed System Expected Output

- The system provides a real-time, GPS-enabled digital dashboard that displays RA values, asset condition, and compliance status across mapped highway segments.
- It generates instant pass/fail alerts for substandard markings, enabling quick identification of critical zones.
- The dashboard also supports data visualization, historical tracking, and predictive maintenance insights, allowing authorities to plan timely repairs and ensure continuous road safety.

4. Innovation — What Makes Us Different

- The proposed system, RetroFusion AI, introduces a hybrid approach for measuring retroreflectivity by integrating AI-based computer vision, photometric sensing with controlled illumination, and retroreflectometer-based calibration into a single or vehicle-mounted platform. Unlike existing solutions that rely either on manual handheld devices or standalone imaging systems, this approach combines the strengths of both hardware precision and AI-driven scalability.
- The innovation lies in multi-modal sensor fusion, where real-time visual detection of road assets is synchronized with accurate light intensity measurements to compute reliable RA values under dynamic conditions such as high-speed movement, low light, fog, and wet surfaces. The inclusion of a calibration mechanism ensures that the system maintains standard-compliant accuracy, addressing one of the key limitations of purely AI-based estimation methods.
- This integrated design enables continuous, non-intrusive measurement at highway speeds, eliminating the need for manual intervention and significantly improving operational safety. Furthermore, the system is built using cost-effective, locally available components, making it highly feasible for large-scale deployment across India’s highway network.

Table 4.1: RetroFusion AI is innovative across distinct dimensions:

Innovation Aspect	Description	Justification (Why it is Innovative)
Hybrid Multi-Modal System	Integration of AI vision, photometric sensors, and retroreflectometer calibration	Overcomes limitations of single-method systems by combining accuracy (hardware) with scalability (AI)
AI-Based Computer Vision	Real-time detection and classification of road markings, signs, studs, and delineators	Eliminates manual identification and enables continuous monitoring at highway speeds
Photometric Sensing with IR Illumination	Measures reflected light intensity under controlled lighting conditions	Ensures accurate RA values in night, fog, and wet conditions where cameras alone fail
Retroreflectometer Calibration	Provides ground-truth validation for measured values	Maintains compliance with IRC/NHA standards and improves measurement reliability
Sensor Fusion Algorithm	Combines AI and sensor data for final RA estimation	Enhances accuracy, reduces errors, and ensures consistency across varying conditions
Vehicle-Mounted Continuous Measurement	System operates while moving at highway speeds	Eliminates need for lane closures and manual intervention, increasing efficiency and safety
Low-Cost, Modular Design	Built using affordable and locally available components	Enables large-scale deployment across India without dependence on expensive imported systems

05. Feasibility

5.1.Complete Hardware BOM & Cost

Table 5.1. Different components with cost (Based on our Survey, all components are available on Amazon India, Robu.in, and Electronicscomp.in.)

Component	Function	Connection	Pi Pin	Approx Cost (INR)
Raspberry Pi 4B (8GB)	Main Controller & AI Processor	—	—	₹8,000
Pi HQ Camera (IMX477)	12.3MP Image Capture	CSI Ribbon	CSI Port	₹5,000
TSL2591 Lux Sensor	IR Light Intensity Measurement	I2C (SDA/SCL)	GPIO 2/3	₹350
u-blox NEO-M9N GPS	Precise Location Tagging ($\pm 1.5\text{m}$)	UART TX/RX	GPIO 14/15	₹1,500
IR LED 850nm (3W)	Controlled IR Illumination	Via MOSFET	GPIO 18	₹300
MOSFET Driver (IRLZ44N)	PWM IR LED Control	Gate Control	GPIO 18	₹100
Retroreflectometer	Ground Truth RA Validation	USB Interface	USB Port	—
Buck Converter (12V $\rightarrow$ 5V)	Stable Power Regulation	—	12V Input	₹300
LiPo Battery (10000mAh)	Portable Power Source	—	12V	₹1,200
Solar Panel	Continuous Charging	—	12V	₹800
Wires, Connectors & Housing	Assembly & Protection	—	—	₹200
TOTAL PROTOTYPE COST				$\approx$ ₹17,750

5.2. Software Stack (All Open Source- free of Cost)

- OS: Raspberry Pi OS (Linux-based, free)
- AI Framework: TensorFlow Lite (edge-optimized, free)
- Computer Vision: OpenCV 4.x (free)
- Sensor Interface: Python 3 + Adafruit CircuitPython (free)
- Fusion Algorithm: Custom Kalman filter implementation (Python)
- Dashboard: Flask/FastAPI backend + React.js frontend (free)
- Cloud: AWS IoT / Firebase (scalable pricing, free tier for prototyping)

6. Scalability & Sustainability

The true test of any infrastructure solution is not whether it works once- it is whether it works everywhere, always, and gets better over time. RetroFusion AI is designed from the ground up for national scale.

Table 6.1. Scalability parameters of our system

Scale Dimension	Capability
Highway Coverage	Works on 2-lane, 4-lane, 6-lane, and 8-lane expressways without reconfiguration
Weather Conditions	Day / Night / Dry / Wet / Foggy — IR illumination ensures weather independence
Speed Range	Effective from 0–120 km/h — highway-speed continuous scanning
Fleet Deployment	Each unit costs ~₹18K; 100 units = ₹18L for all-India coverage
AI Model Improvement (GEMMA 4)	Cloud-aggregated data trains a continuously improving national road quality model
Drone Extension	Architecture supports drone-mounted variant for gantry signs and inaccessible areas
Integration	REST APIs for direct integration with NHAI's existing project management systems

A fleet of 50 RetroFusion AI units, each covering 500 km/day, can survey NHAI's entire highway network in under 6 days versus months with current methods(manual). This is not an upgrade. It is a revolution.

6.1. Competitive Comparison

Table 6.2: Competitive Comparison

Parameter	Handheld Device (Current)	Imported Vehicle System	RetroFusion AI+Pro
Measurement Speed	12–20 per hour	High — but costly	Continuous at highway speed
Operator Safety	⚠ Dangerous	Safe (in-vehicle)	Safe (in-vehicle)
AI/ML Analysis	None	Limited	Full CNN + fusion
GPS Tagging	No	Yes	Yes (±1.5m)
Weather Coverage	Day, dry only	All conditions	All conditions (IR)
Cost per Unit	₹2–5 lakh (device)	₹50–200 lakh (vehicle)	~₹18,000 (prototype)
Indian Availability	Imported	Imported	100% Made in India
Cloud Dashboard	No	Proprietary (costly)	Open-source, free
Scalability	Very Low	Low (cost barrier)	Very High

6.2 Environmental Sustainability

The system's optional solar panel charging makes it fully energy-independent during daytime operation. The lightweight hardware (under 2 kg) adds negligible load to any inspection vehicle. No hazardous materials are used, and the system's longevity reduces the need for frequent replacement, a net positive for lifecycle sustainability.

SDG Alignment



The proposed system aligns primarily with **SDG 3 (Good Health and Well-being)** by reducing road accidents and further supports **SDG 9 and SDG 11** through innovation-driven infrastructure monitoring and safer transportation systems.

7. Implementation Roadmap

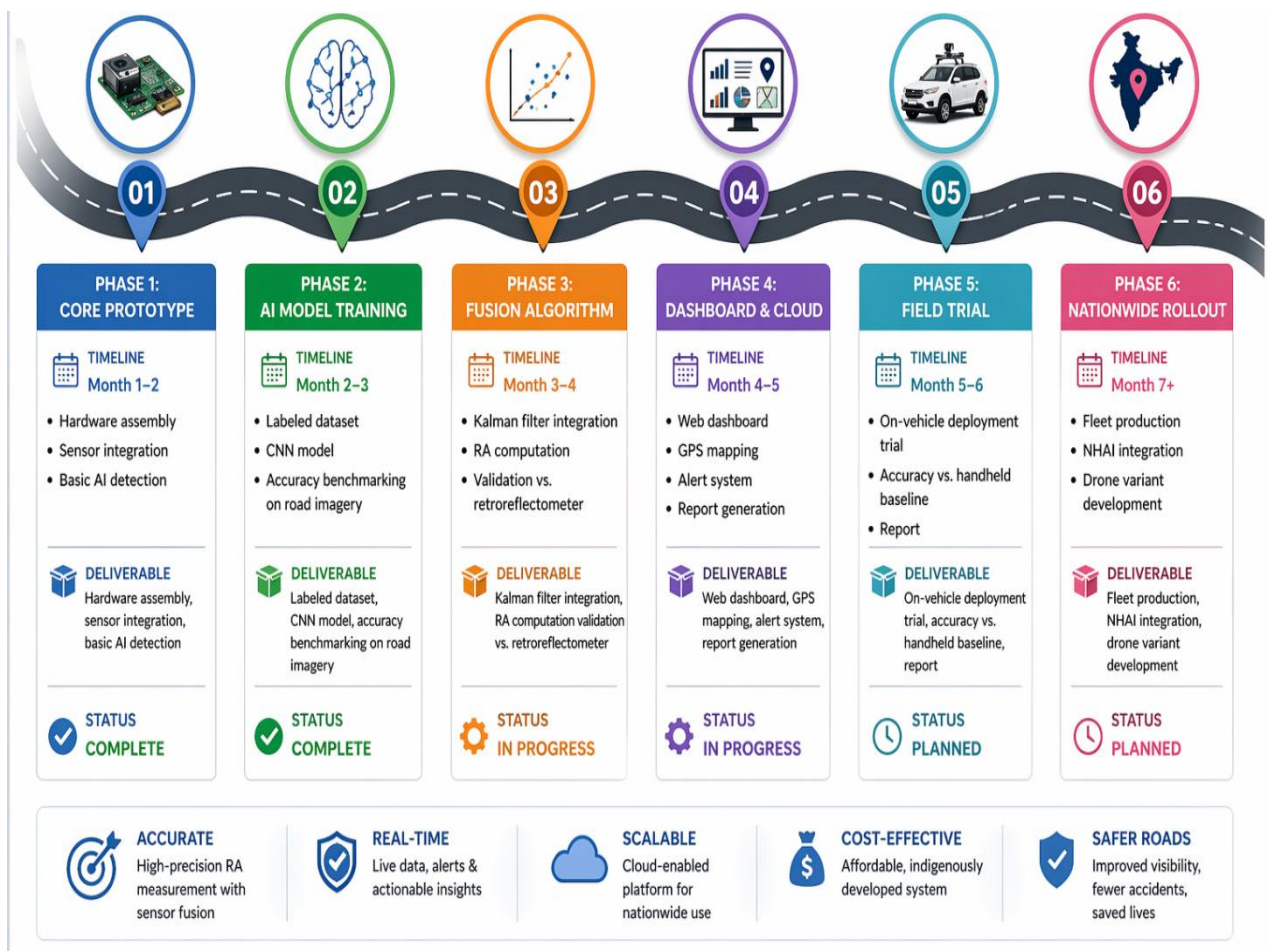


Fig: 7.1 : Implementation Roadmap

8. Future Scope & Vision

RetroFusion AI is not just a hackathon prototype, but it is the seed of a national road intelligence platform. With NHAI's support, here is where this technology leads:

- **Drone-Mounted Variant:** For gantry-mounted signs and areas inaccessible to ground vehicles, UAV integration using the same AI core with a lighter sensor payload.
- **Predictive Maintenance AI:** Machine learning models that forecast when specific highway sections will fall below minimum RA thresholds- based on traffic load, weather patterns, and material type- enabling pre-emptive maintenance scheduling.
- **National Road Quality Index:** Aggregated retroreflectivity data from the fleet creates a live, publicly accessible National Road Quality Index - enabling transparent accountability for highway contractors.
- **Multi-Parameter Expansion:** The same platform architecture can be extended to measure road surface roughness (IRI), pothole detection, lane marking fading index, and sign legibility scores - transforming the system into a comprehensive Highway Health Monitor.
- **OEM Integration:** Partnership with Ashok Leyland, TATA Motors, and highway inspection vehicle OEMs to offer RetroFusion AI as a factory-fitted option for NHAI fleet vehicles.

8. Results/Expected Output

The proposed system delivers **real-time, accurate, and actionable outputs for highway authorities:**

- **RA Values (Retroreflectivity Measurement):** Continuous measurement of retroreflectivity (RA values) for road markings, traffic signs, studs, and delineators under real driving conditions.
- **Pass/Fail Alerts:** Instant classification of assets based on compliance with IRC/NHAI standards, highlighting non-compliant or degraded elements for immediate attention.
- **GIS Mapping:** All measurements are GPS-tagged and visualized on a geospatial dashboard, enabling easy identification of critical zones across highway segments.
- **Maintenance Suggestions:** Data-driven insights provide prioritized maintenance recommendations, helping authorities plan timely repainting, replacement, and resource allocation.

9. Conclusion

RetroFusion AI transforms highway safety from a periodic checkbox into a continuous, intelligent, data-driven process — making India's roads safer for every driver, every night.

RetroFusion AI stands out as an **integrated solution** by combining **accuracy, speed, safety, and scalability** into a single platform. Its innovative fusion of AI and hardware enables **continuous, real-time monitoring**, eliminating the risks and limitations of manual methods.

The solution is **fully aligned with NHAI requirements**, addressing vehicle-based measurement, AI-driven analysis, and their integration into a unified system. With its **low-cost, deployable design and high-impact outcomes**, it offers a practical pathway toward **safer highways, improved compliance, and data-driven infrastructure management** across India.

RetroFusion AI | Safer Highways Through Intelligent Vision

Our Details

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Annexures :

Website :

https://antigravity-retrofusion-o956is25nqjtrsbudad6ja.streamlit.app/Live_Monitor

Simulation of data flow:

[RetroFusion AI+ Pro — Interactive 3D Hardware Simulation | Claude](#)